

Search, Read, Cite

Building a Reliable Research Assistant Agent

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Why Source-Ground Research Assistants Matter

LLMs are useful for research, but reliable answers require verified evidence, exact quotes, and citations

The Problem

Current LLM gap

- Answers may be factually incorrect
- Claims may be unsupported
- Citations can be missing or fabricated
- Complex questions often require multiple sources

Why It Matters

Research workflows

- Students needed cited explanations
- Researchers need paper summaries and quote extraction
- Policy / business users need current, source-based briefs

Our Project Goal

Search -> Read -> Cite

- Build a tool-using research assistant
- Use web search, Wikipedia, PDF parsing, quote search and citation tools
- Compare ReAct and CoT on tool choice, coverage, runtime, and completion

Research Questions

We evaluate both the final answer and the process that produced it

RQ1: Agent strategy

How does a ReAct-style loop compare to single-pass CoT?

Hypothesis: ReAct improves factuality and tool selection but increases latency/cost.

RQ2: Tool diversity

How does the number and diversity of tools affect task completion and reasoning efficiency?

Hypothesis: More tools improve coverage but can introduce selection errors.

RQ3: Verified Quote-retrieval system

Can verified quote retrieval outperform generic web search for citation-ready evidence?

Hypothesis: Verified quote retrieval return more real, verified quotes, and claim-level evidence than generic web search

Scope boundary

We focus on source-grounded research QA with 10 curated benchmark questions. We do not train a new LLM or support unlimited web browsing. Full citation verification remains future work.

System Architecture: Search -> Read -> Cite

The agent turns a user question into a sourced answer through tool-mediated steps



Tool Suite: Search, Read, Quote, and Cite

Each tool handles a specific failure point in source-grounded research QA

1. Search Tools

- **Web Search** – Find current and domain-relevant sources.
- **Wikipedia API** – Provide background context

2. Reading Tools

- **PDF Fetcher + Parser** – Extract text from academic papers and reports; OCR supports scanned/image-only PDFs
- **Document Summarizer** – Summarize or answer questions about the uploaded / active document

3. Evidence Tools

- **Quote Search** – Retrieve verified exact quotes from the active document with character-level locations
- **Paper Quote Finder** – Search external papers / sources and return supporting quotes for user-provided claims

4. Citation Tools

- **Citation Generator**– Create formatted source attribution without guessing missing metadata

Tools are modular: the agent selects the smallest tool chain needed for the research task

Finding Paper Quotes Pipeline



Why our pipeline beats generic web search:

- **Finds real quotes**, not snippet guesses
- **Returns verified quotes** matched to source text
- **Provides claim-level evidence**, not just plausible URLs
- **Produces citation-ready support** for research writing

Agent Strategies Compared

We compare different ways of deciding when and how to use tools

Single-pass CoT

One model call

Prompt includes instructions to reason and cite

Fast baseline but limited interaction

Zero-shot function calling

Model directly emits tool calls

Structured function schemas

Good for argument accuracy

ReAct-style loop

Interleaves reasoning, action, observation

Can recover from failed searches

Slower but more adaptive

Benchmark Dataset

10 research questions with expected evidence and tool-use behavior

Question types

- Current factual QA
- Multi-hop comparison
- Paper-specific extraction
- Definition + evidence synthesis
- Citation verification

Gold annotations

- Reference answer
- Acceptable sources
- Expected tool sequence
- Required citations
- Difficulty label

Domains

- AI / ML
- Science
- Policy
- Business
- Education
- General knowledge

Quantitative Evaluation Metrics

The scorecard separates answer quality, tool behavior, and system performance

Answer Quality

Factual correctness, completeness, groundedness

Tool Use

Tool selection accuracy, argument correctness, task completion

Citation Quality

Citation precision/recall, source support, formatting

Efficiency

Latency, number of calls, tokens, cost

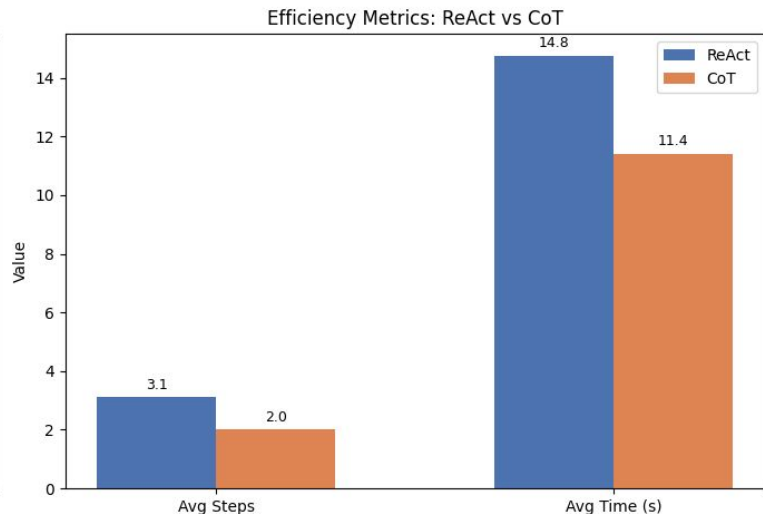
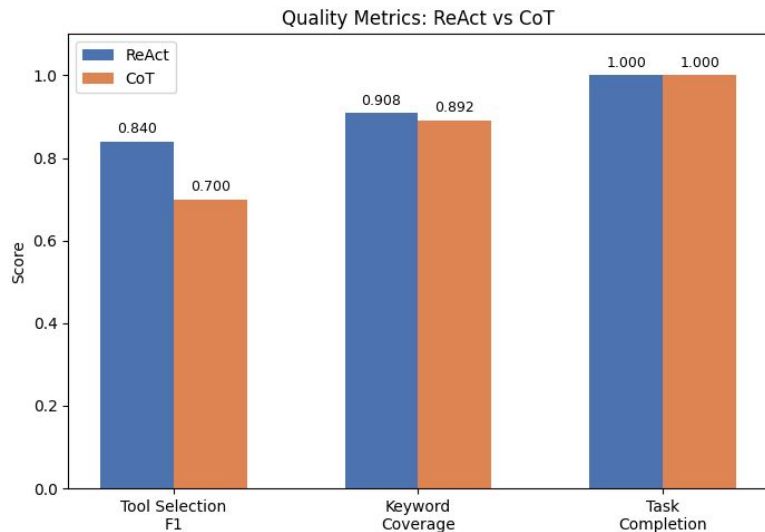
Retrieval

Hit Rate@k, MRR, NDCG where gold evidence exists

Robustness

Failure rate, invalid calls, repeated calls, hallucinated evidence

Quantitative Results: ReAct vs. CoT



Key takeaway:

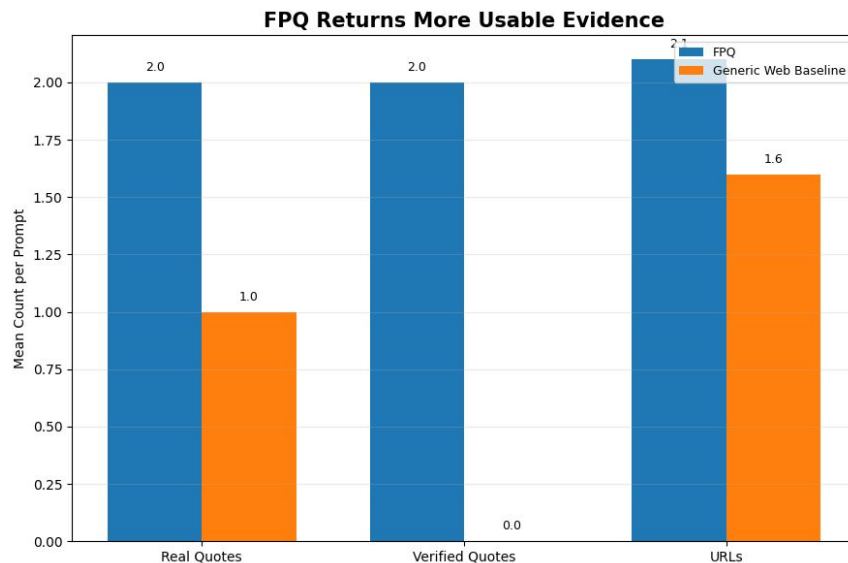
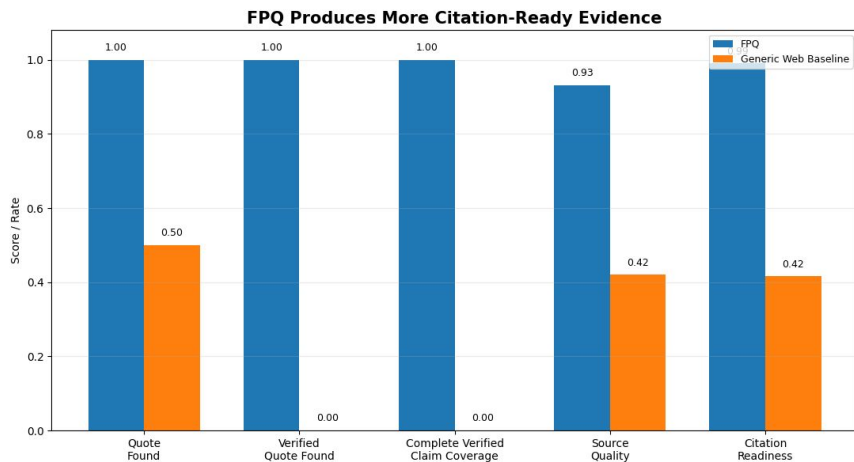
- ReAct = better tool choice
- CoT = faster and cheaper

Key Findings: ReAct vs. CoT

Metric	ReAct	CoT
Avg Tool Selection F1	0.840	0.700
Avg Keyword Coverage	0.908	0.892
Avg Steps	3.1	2.0
Avg Time (seconds)	14.8	11.4
Task Completion Rate	1.000	1.000

- Both strategies achieve 100% task completion and similar keyword coverage
- ReAct selected tools more accurately, especially for specialized tools like `quote_search`
- CoT is 30% faster with fewer steps, offering a better cost-quality tradeoff for straightforward questions

Finding Paper Quotes (FPQ) Results



Key Findings:

- FPQ significantly outperforms generic web baselines across all citation-readiness metrics.
- The specialized quote search tool produces 2x more usable evidence compared to standard search methods.
- Verified Quote Found and Source Quality show the most dramatic improvements in the FPQ pipeline.

Qualitative Evaluation and Error Analysis

We identify where the research process breaks down

Agent chooses the wrong next tool or loops unnecessarily

Example: uses `web_search` for a quote request instead of `quote_search`

Search returns irrelevant or incomplete sources

Example: retrieves a general article when the task needs the original paper

Parser extracts incomplete, noisy, or no text

Example: scanned PDF has no extractable text, so quote search cannot run

Answer uses retrieved sources but draws unsupported conclusions

Example: source mentions RAG, but answer over claims that RAG fully eliminates hallucination

Citation exists but does not actually support the claim

Example: cites the right paper but the quoted passage is unrelated to the sentence

Ablation: Effect of Quote Search Tool

Question	Setting	Tool called	Steps	Keyword Coverage
q04	With quote_search	quote_search	1	1.00
	Without quote_search	web_search, fetch_pdf, web_search, fetch_pdf	4	1.00
q09	With quote_search	quote_search	1	0.33
	Without quote_search	Web_search, web_search, fetch_df, fetch_pdf	4	0.33

- Removing quote_search forces the agent to use 4x more tools calls to attempt the same task
- The key difference is qualitative: only quote_search provides verified exact-substring quotes with character locations; general tools can only paraphrase

Major Contributions and Practical Impact

Our project contributes both a working research agent and an evaluation framework for reliable GenAI research workflows.

1. Working Research agent

Built a demo that searches live sources, reads web/PDF content, retrieves quotes, and generates cited answers.

2. Evaluation Framework

Created a 10-question benchmark and compared ReAct and CoT using tool selection, coverage, steps, runtime, and completion.

3. Trust and Efficiency

Users can inspect sources and tool logs, while the system automates the search-read-cite workflow.

4. Scalability and Learning Value

The modular design supports future tools, and the step-by-step logs make the system easier to debug and evaluate.

Limitations and Future Work

The project is scoped for a semester, so we explicitly track what remains open

Limitations

- Benchmark contains **only 10 questions** which limits the statistical power – Mann-Whitney test could not detect significance even where trends were visible
- The questions cover various task types, but they are still manually curated and may not fully represent real-world research workflows
- Text-based PDFs work, but scanned or image-only PDFs may fail because they do not contain extractable text.
- Find paper quotes depends on web availability – paywalled, unpublished, or non-indexed sources can't be retrieved
- The agent generates citations and track sources, but not yet have a separate verifier that checks whether sentence-level claim is actually supported by the cited evidence

Future work

- **Expand the benchmark** – A larger benchmark with more domains, more difficulty levels, and more adversarial examples.
- **Multi-document upload** – allow users to query across a library of uploaded PDFs
- **Add OCR** – support for scanned PDFs and image-only documents so the system can process wider range of materials
- **Human evaluation study** – Ask annotators to rate factual correctness, citation support, usefulness, and trustworthiness.
- **Add citation-level verification** – Build a verifier that checks each final-answer claim against the retrieved evidence and flags unsupported claims before output

Thanks for listening!

Reliable research assistants should be evaluated by both answer quality and research process quality

Search

Find the right evidence

Read

Extract and reason over sources

Cite

Tie claims to verifiable support

Github Repository: <https://github.com/IneshV/GenAI-LLM-Final-Project>